

Answer Key & Rationales

Basic Revision Quiz: Mechanical Oscillations

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Quick Answer Grid (Final Choices)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	A	D	C	C	C	B	B	D	C

Detailed Rationales & Point Breakdown (10.0 pts Total)

- **Question 1 (B) [0.5 pt]:** Hooke's Law states that the restoring force of an ideal spring is directly proportional to its elongation, but always acts in the opposite direction of the displacement. Thus, $\vec{F} = -kx\vec{i}$.
- **Question 2 (A) [1.5 pts]:** The proper angular frequency is calculated using the formula $\omega_0 = \sqrt{\frac{k}{m}}$.
Substituting the given values: $\omega_0 = \sqrt{\frac{10}{0.1}} = \sqrt{100} = \mathbf{10 \text{ rad/s}}$.
- **Question 3 (D) [0.5 pt]:** Applying Newton's Second Law to an undamped horizontal spring yields $-kx = ma$. Since $a = x''$, this rearranges to $mx'' + kx = 0$, dividing by m gives the standard differential equation $x'' + \frac{k}{m}x = 0$.
- **Question 4 (C) [1.5 pts]:** The maximum speed is given by the derivative amplitude: $V_{max} = \omega_0 X_m$.
First, convert the amplitude to standard SI units (meters): $X_m = 5.0 \text{ cm} = 0.05 \text{ m}$.
Calculation: $V_{max} = (20 \text{ rad/s}) \times (0.05 \text{ m}) = \mathbf{1.0 \text{ m/s}}$.
- **Question 5 (C) [1.0 pt]:** The kinetic energy is maximum when the speed is maximum. The object moves fastest as it passes through the **equilibrium position** ($x = 0$), where all the potential energy has been converted to kinetic energy.
- **Question 6 (C) [1.5 pts]:** Because the system is undamped (conserved), the total mechanical energy is equal to the maximum elastic potential energy stored at the extremes.
 $ME = EPE_{max} = \frac{1}{2}kX_m^2 = 0.5 \times (200 \text{ N/m}) \times (0.2 \text{ m})^2 = 100 \times 0.04 = \mathbf{4.0 \text{ J}}$.
- **Question 7 (B) [1.0 pt]:** By definition of Simple Harmonic Motion, the acceleration is directly proportional and opposite to the position, given by the relation $\mathbf{a} = -\omega_0^2 \mathbf{x}$.
- **Question 8 (B) [0.5 pt]:** Mechanical energy is strictly conserved only in an isolated system where no non-conservative forces (like friction or air resistance) do work. This implies the surface must be **frictionless**.

- **Question 9 (D) [1.0 pt]:** In a pseudo-periodic regime, the friction force constantly does negative work on the system. This dissipates mechanical energy (as heat), which directly results in a continuous **decrease in the amplitude** of the oscillations.
- **Question 10 (C) [1.0 pt]:** The kinetic energy and potential energy of an oscillator complete two full cycles during a single cycle of position (they hit their maximums twice per position period). Thus, $T_{energy} = \frac{T_0}{2} = \frac{2.0}{2} = 1.0 \text{ s}$.